

SmartBiz Pro ERP

Funding Proposal & Full Project Plan

Date: 2026-01-13

Document Version: 2.0

Prepared for: Investors / Donors / Strategic Partners

Table of Contents

1. Executive Summary
2. Problem Statement & Market Gap
3. Solution Overview
4. Product Scope: Modules & Features
5. Predictive Analytics
6. Offline-First Architecture & Conflict Resolution
7. Technical Architecture
8. API Specification Examples
9. Roadmap & Milestones
10. Team & Governance
11. Budget & Use of Funds
12. Appendices (Wireframes)

SmartBiz Pro ERP — Funding Proposal & Full Project Plan

****Document Version:**** 2.0

****Date:**** 2026-01-13

****Prepared for:**** Investors / Donors / Strategic Partners

****Prepared by:**** SmartBiz Pro Development Team

1. Executive Summary

SmartBiz Pro is a modern, modular ****offline-first ERP**** designed for multi-industry SMEs and growing enterprises. It provides end-to-end business management (Finance, Inventory, Procurement, HR/Payroll, Production/Projects) while maintaining ****full operational continuity during internet outages****.

****Funding ask (baseline):**** ****USD \$235,750**** for a 10-month build (6–8 core team), plus go-to-market pilot costs.

2. Problem Statement & Market Gap

2.1 Key pain points in existing ERP solutions

- ****Connectivity dependence:**** Cloud-only ERPs degrade when internet fails; frontline operations stall (POS, warehouse, field teams).
- ****High integration complexity:**** Data compatibility, customization, and integration costs remain key barriers.
- ****Implementation failure risk:**** Many ERP programs run over budget/timeline due to change management, data migration, training, and scope creep.

2.2 Opportunity

SmartBiz Pro targets the underserved segment of businesses that need:

- Reliable operations ****offline**** for extended periods
- Local compliance (VAT/PAYE and audit controls)
- Faster deployment via industry templates
- Actionable insights (predictive analytics) rather than static reports

3. Solution Overview (SmartBiz Pro)

3.1 Core Value Proposition

1. **Offline-first** transaction processing with robust sync and conflict resolution
2. **Composable modular ERP**: deploy only what you need; scale as you grow
3. **Industry Quick-Start Templates**: go-live in hours, not weeks
4. **Predictive Analytics**: business health score, forecasts, anomaly alerts
5. **API-first platform**: integrations with payments, banks, eCommerce, SMS/WhatsApp

3.2 Target Industries

Retail/Wholesale, Healthcare/Clinics, Schools/Education, NGOs/Projects, Manufacturing.

4. Product Scope: Modules & Detailed Features

4.1 Production / Project Management

- Multi-project hierarchies and budgets
- Resource allocation, capacity planning, utilization dashboards
- Milestones and Gantt timelines
- Material requirements integrated with Inventory (MRP-lite)
- Collaboration: comments, attachments, mentions

4.2 Stakeholders (Customers/Suppliers)

- Custom fields, segmentation, credit limits, payment terms
- Interaction history and communication logs
- Customer portal: invoices, statements, payments
- Supplier performance scoring (lead time, quality, price variance)
- Bulk import/export

4.3 Budget Management

- Budget codes and hierarchies
- Approval workflows for estimates
- Variance analysis and alerting
- Rolling forecasts and what-if scenarios

4.4 Payroll Management

- Multiple payroll schedules; payslips; bank vouchers
- PAYE, statutory deductions, allowances & benefits
- Loans and advances

- Year-end certificates
- Electronic distribution (email/SMS)

4.5 Store / Inventory Management

- GRN automation; stock ledger movements (append-only)
- FIFO/LIFO/Weighted valuation
- Barcode scanning; batch/expiry
- Multi-location & transfers
- Cycle counts and audit trails
- Reorder automation + forecasts

4.6 Account / Cash Management

- Multi-account statements; transfers; petty cash
- Bank reconciliation; cashflow forecasting
- Multi-currency support
- Audit trail and financial ratio dashboards

4.7 Purchases Management

- Requisition → approvals → PO → GRN → invoice (3-way match)
- VAT computation, spending analytics
- Contract & price list management

4.8 HR Management

- Employee records + document storage
- Leave workflows; attendance/time tracking
- Performance appraisals; training records
- Org chart and recruitment workflow

4.9 Dashboards & Analytics

- Executive dashboard with KPIs
- Custom report builder
- Scheduled reports
- Predictive insights embedded in workflows

4.10 System Administration

- RBAC, permissions templates
- Audit logging, activity monitoring

- Backup & recovery, configuration

5. Predictive Analytics: How It Enhances ERP Value

SmartBiz Pro upgrades ERP from **recording** to **advising**.

5.1 High-ROI Predictive Use Cases

- **Cash runway & cashflow forecast** (7/30/90 days)
- **AR late-payment risk** scoring per customer
- **Stockout prediction** and reorder recommendations
- **Budget overrun prediction** based on burn rate
- **Anomaly detection** for fraud/leakage (stock adjustments, duplicate payments)

5.2 Delivery Strategy (Phased)

- **Phase A (MVP):** rules + moving averages + SQL-based forecasts
- **Phase B:** lightweight ML scoring service (risk/churn)
- **Phase C:** anomaly detection + recommendation engine

6. Offline-First Architecture & Conflict Resolution

6.1 Client-side Offline Components

- Service Worker caching + background sync
- IndexedDB (Dexie) as local database
- Local outbox queue (operations log)
- Sync engine (push/pull + prioritization)
- Conflict resolution UI for manual review

6.2 Conflict Resolution Strategy

Not all data conflicts are equal:

- **Finance** (invoices/payments/ledger): no silent merges; manual review
- **Inventory:** event-based append-only stock movements merge safely
- **Master data:** field-level merge with safeguards

6.3 Key Mechanisms

- **Idempotency keys** per operation to prevent duplicates

- ****Optimistic concurrency control**** using record versioning
- ****Conflict records**** stored and surfaced to users

6.4 Simplified Sync Pseudocode

```
async function syncOfflineChanges() {  
  // 1. Connectivity check  
  if (!navigator.onLine) return;  
  
  // 2. Prioritized queue  
  const ops = await db.outbox  
    .where('status').equals('PENDING')  
    .sortBy('priority');  
  
  const batch = ops.slice(0, 200);  
  
  // 3. Push operations (idempotency keys)  
  const res = await api.post('/sync/push', {  
    deviceId,  
    operations: batch  
  });  
  
  // 4. Mark synced + store conflicts  
  for (const ok of res.applied) {  
    await db.outbox.update(ok.operationId, { status: 'SYNCED' });  
  }  
  
  for (const c of res.conflicts) {  
    await db.outbox.update(c.operationId, { status: 'CONFLICT' });  
    await db.conflicts.put(c);  
  }  
  
  // 5. Pull latest server changes  
  await pullLatestChanges();  
}
```

7. Technical Architecture

7.1 High-level Architecture

Client (PWA/Mobile/Desktop)

- Next.js PWA, Service Worker
- IndexedDB + Sync Engine

|

| HTTPS/WebSocket

```

v
API Gateway (Hono on Bun)
- Auth, RBAC, Rate-limits
- OpenAPI docs, webhooks
|
v
Services (modular)
- Finance, Inventory, HR/Payroll, Projects, Reporting
|
v
Supabase (Postgres + Auth + Realtime + Storage)
- RLS policies
- Audit logs

```

7.2 Technology Stack

- **Frontend:** Next.js (React), Tailwind, shadcn/ui, React Query, Zustand
- **Offline:** Dexie (IndexedDB), Service Worker, background sync
- **Backend:** Bun runtime + Hono, Zod, Drizzle ORM, Pino logging
- **Database:** Supabase Postgres + RLS; Redis cache; object storage
- **DevOps:** Docker, GitHub Actions, Kubernetes (prod), Prometheus/Grafana, Sentry

8. API Specification (Examples)

8.1 REST Endpoints (sample)

```

openapi: 3.0.3
info:
  title: SmartBiz Pro API
  version: 1.0.0
paths:
  /auth/login:
    post:
      summary: Login
  /inventory/items:
    get:
      summary: List items
    post:
      summary: Create item
  /inventory/stock-movements:
    post:
      summary: Create stock movement (GRN/SALE/ADJUST)
  /finance/invoices:

```


get:
summary: List invoices
post:
summary: Create invoice
/sync/push:
post:
summary: Push offline operations
/sync/pull:
get:
summary: Pull server changes since cursor

9. Roadmap & Milestones (10 Months)

Phase 1 (Weeks 1–8): Foundation

- Setup monorepo, CI/CD, environments
- Auth + RBAC + tenanting
- Offline engine v1 + sync logs
- UI framework + dashboard skeleton

Phase 2 (Weeks 9–20): Financial Core

- Stakeholders + Budgets
- Cash/Bank + Purchases
- Store/Inventory (offline-first)

Phase 3 (Weeks 21–32): HR & Advanced

- HR + Payroll compliance
- Production/Projects
- Cross-module integration tests

Phase 4 (Weeks 33–40): Industry packs + Polish

- Templates for 3 industries
- Predictive analytics v1
- Security audit + performance optimization

10. Team & Governance

- PM (1), Tech Lead (1), FE (2), BE (2), DevOps (1), QA (1)

- Agile 2-week sprints, demo every sprint
- Change-control board to prevent scope creep

11. Budget & Use of Funds (Baseline)

Category	Total (USD)
Personnel (10 months, 6–8 team)	190,000
Infrastructure	10,000
Tools & Services	5,000
Contingency (15%)	30,750
Grand Total	235,750

12. Appendices

A. Wireframes

- `wireframe\_dashboard.png`
- `wireframe\_pos.png`
- `wireframe\_inventory.png`
- `wireframe\_conflicts.png`

Appendix: Wireframes

Executive Dashboard

Wireframe: Executive Dashboard

Sidebar - Modules - Favorites - Recent - Support

Top KPI Strip Cash • Revenue • Expenses • Stock-outs • Payroll Due

Cashflow Forecast (7/30/90 days) Alerts & Recommendations

Business Health Score Risk signals (Late payers, Dead stock, Budget burn)

Recent Transactions Sales • Purchases • Payments

Approvals Queue PR • PO • Expense • Leave

SmartBiz Pro ERP • Wireframe (concept) • 2026-01-13

Retail POS

Wireframe: Retail POS (Offline-First)

Product Search Barcode/QR Categories	Cart Items • Qty • Price • Discounts Offline badge + sync status	Customer Select / Add Credit limit Loyalty
	Payment Cash • Mobile Money • Card Split payments	Receipt Print / SMS / WhatsApp Tax Invoice

SmartBiz Pro ERP • Wireframe (concept) • 2026-01-13

Store & Inventory

Wireframe: Store & Inventory

Actions GRN Issue Transfer Counts Adjust	Inventory Overview Stock by location • Reorder alerts • Expiry	
	Items Table Search • Filters Batch/Expiry Offline edits queued	Reorder Recommendations Forecasted demand Supplier lead time Generate PO

SmartBiz Pro ERP • Wireframe (concept) • 2026-01-13

Conflict Resolution

Wireframe: Offline Sync Conflict Resolution

Conflicts List Invoice # Payment Customer Stock	Local Version Fields highlighted Created offline	Server Version Latest state Audit trail
	Resolve: Use Server • Keep Mine • Merge Fields • Create Adjustment • Send for Review	